

Adaptive Radio Resource Management for Wi-Fi Networks Using Constrained Bayesian Optimization

Abstract—This study presents a Radio Resource Management (RRM) framework that combines the predictive efficiency of constrained Bayesian Optimization (BO) with the stability of IEEE-standard RRM configuration. The system operates in two distinct phases—an intensive offline exploration phase and a continuous online deployment phase—linked through a A/B toggle choice. The architecture uses a Gaussian Process (GP)-driven Bayesian optimizer implemented in Python using Optuna, connected with MATLAB for realistic online evaluation. Budget enforcement, cool-off intervals, and stability controllers ensure compliance with Quality of Service (QoS) and regulatory limits, enabling smart data-driven Wi-Fi management.

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I. INTRODUCTION

Modern Wi-Fi environments are characterized by dense deployments and fluctuating interference patterns, demanding adaptive RRM techniques. Conventional methods—such as Cisco's TPCv1 algorithm focuses on reducing power consumption while trying to maintain a high throughput. While stable, this take a longer time to converge and cannot adapt effectively to changes in real time.

In contrast, the proposed system learns from live network behavior using Bayesian Optimization (BO). It adaptively adjusts:

- Transmit Power (P_{TX}),
- OBSS-Preamble Detection Threshold ($OBSS_{PD}$), and
- Channel Bandwidth (W_{CH}),

to maximize median throughput (P_{50}) while maintaining Retry Rate and EIRP constraints.

II. PROBLEM FORMULATION

The RRM task is modeled as a constrained black-box optimization:

$$\max_{\mathbf{x}} \mathcal{T}_{P50}(\mathbf{x}) \quad (1)$$

$$\text{s.t. } \mathcal{R}_{P95}(\mathbf{x}) \leq 8\%, \quad (2)$$

$$(P_{TX} + G_{ANT}) \leq 30 \text{ dBm}, \quad (3)$$

The following variables are used to define the RRM objective function and control parameters:

$\mathcal{T}_{P50}$ **P50 Throughput:** Represents the 50th percentile of client downlink throughput, which is the performance metric being maximized.

$\mathcal{R}_{P95}$ **P95 Retry Rate:** Represents the 95th percentile of the MAC retry rate, which is the stability constraint being minimized.

P_{TX} **Radio Transmitter Power:** The power output by the transmitter radio $\mathbf{x}$.

G_{ANT} **Antenna Gain:** The efficiency and directivity of the antenna, a physical parameter used in calculating EIRP.

Optimization Variables and Ranges

- P_{TX} : 17 to 30 dBm (hardware and EIRP-limited),
- $OBSS_{PD}$: -82 to -62 dBm (spatial reuse aggressiveness),
- W_{CH} : $\{20, 40\}$ MHz.

Each parameter interacts nonlinearly with interference, SINR, and retry rates—making the problem well-suited for Gaussian Process-based optimization modeling.

III. CISCO'S TPCv1 ALGORITHM—TRADITIONAL APPROACH

The TPCv1 (Transmit Power Control version 1) algorithm operates in a two-stage process to adjust a radio's transmit power to its ideal level (Tx_{Ideal}). The primary objective is to ensure the radio's coverage is optimized, preventing both excessive power (which causes interference) and insufficient power (which reduces performance).

A. The Calculation

The core of the algorithm is a straightforward calculation that determines Tx_{Ideal} based on three key parameters:

$$Tx_{\text{Ideal}} = Tx_{\text{Max}} + (TPCv1_{\text{Threshold}} - RSSI_{3\text{rd}})$$

- 1) Tx_{Max} : This is the maximum supported transmit power for the radio being used.
- 2) $TPCv1_{\text{Threshold}}$: This is the power threshold. It represents the noise floor or desired signal strength from neighboring Access Points (APs).
- 3) $RSSI_{3\text{rd}}$: This is the Received Signal Strength Indicator of the third loudest AP found in the current AP's neighbor list. By focusing on the third loudest neighbor, the algorithm ensures that the AP is not completely isolated but also prevents power from being reduced too drastically based on an overly close neighbor. Crucially, this is based on the transmit power of neighbors, not the receive power of clients.

B. Power Adjustment Logic

Once Tx_{Ideal} is calculated, it is compared with the radio's current transmit power (Tx_{Current}) and decision is based on :

- If Tx_{Ideal} is higher than Tx_{Current} , then power is increased. The AP can safely transmit at a higher level without causing excessive overlap.
- If Tx_{Ideal} is lower than Tx_{Current} , then power is decrease. The AP is transmitting too loudly relative to its neighbors, and its power should be reduced to minimize interference.

IV. SYSTEM ARCHITECTURE-ADAPTIVE APPROACH

The proposed RRM framework adopts a state-driven architecture inspired by adaptive control loops. Each phase corresponds to a distinct stage in the optimization and deployment lifecycle, ensuring structured progression. Every configuration update is validated against operational constraints, stability heuristics, and real-time client feedback to maintain robustness and reliability.

A. Offline Simulation: Exploration Phase

The algorithm starts with an offline exploration phase, during which a Bayesian optimizer-Gaussian Process sampler—executes $N = 70$ trials within a deterministic Python-Script. Each trial outputs a new configuration vector:

$$\mathbf{x} = [P_{\text{TX}}, \text{OBSS}_{\text{PD}}, W_{\text{CH}}]$$

and computes its reward based on predicted throughput and retry statistics. **This pre-training stage establishes a robust initial model that characterizes the overall performance space**, thereby allowing the optimizer to initiate online deployment from a stable configuration.

B. Time-Slotting of the Bayesian Optimization Study

To ensure that the optimization framework adapts effectively to daily variations in network load, interference, and ambient noise, the Bayesian Optimization (BO) process is divided into **eight fixed time slots per day**, each spanning three hours. **Every slot runs an independent Optuna study with its own substitute model and historical trial database.** This temporal partitioning enables the optimizer to learn *time-localized channel behavior*, capturing fluctuations associated with user-density shifts, congestion cycles, and stochastic environmental noise. By isolating studies across consistent temporal windows, the approach mitigates cross-period interference, reduces model drift, and enhances stability while remaining responsive to short-term variations. Each per-slot study thus results as a targeted statistical model trained to the noise and traffic characteristics specific to that time interval.

To further stabilize objective evaluations, the instantaneous P50 throughput is *smoothed before being consumed by the BO reward*. A fixed-length deque (size 3) stores the three most recent P50 measurements, and the smoothed value used at time t is computed as

$$\widetilde{P50}_t = \frac{1}{3} (P50_t + P50_{t-1} + P50_{t-2}). \quad (4)$$

Thus, the raw $P50_t$ is replaced with $\widetilde{P50}_t$ prior to computing the reward. This smoothing attenuates transient fluctuations caused by fading, bursty interference, and short-lived client dynamics, thereby reducing variance in the objective evaluations passed to the Gaussian-process surrogate. The combined effect of the time-slotting scheme and throughput smoothing yields a more stable BO trajectory, prevents over-reaction to outlier measurements, and promotes reliable convergence toward robust RRM configurations.

C. Gaussian Process Sampler

The optimization framework employs Optuna's **Gaussian Process (GP)** sampler to model the nonlinear relationship between the RRM control parameters— P_{TX} , OBSS_{PD} , W_{CH} —and the resulting network performance metrics. Given the low-dimensional configuration space, the GP offers highly detailed interpolation with inherent uncertainty quantification, thereby resulting in an efficient exploration–exploitation balance. Unlike the **Tree-structured Parzen Estimator (TPE)** and Random samplers, the GP sampler captures global parameter correlations, enhancing data efficiency and convergence stability. Moreover, in contrast to the **Covariance Matrix Adaptation Evolution Strategy (CMA-ES)**, it provides predictive confidence bounds that enable safer, uncertainty-aware parameter updates. The GP sampler is further preferred for its native support of constraint functions, permitting the direct incorporation of safety and performance limits within the optimization process. These characteristics render it particularly effective for resource-constrained and safety-critical RRM optimization tasks.

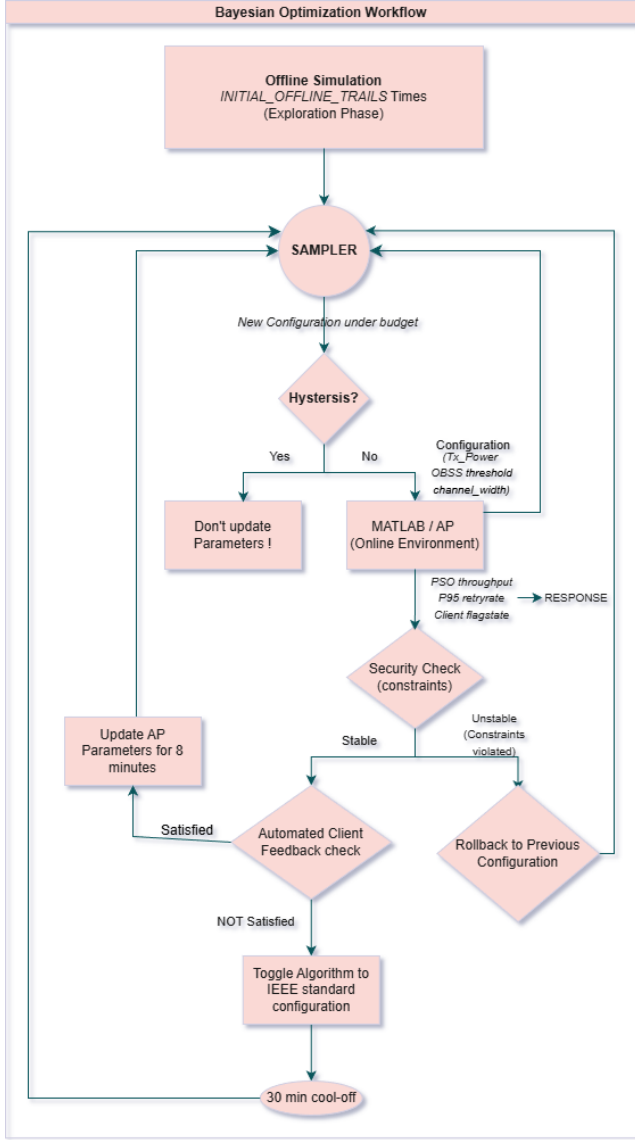


Fig. 1: State-based architecture derived from the Bayesian RRM control diagram.

D. Hysteresis-Based Control for Stable Convergence

A **hysteresis-based control mechanism** was incorporated into the Bayesian optimization framework to reduce **oscillatory parameter** updates and promote stable and better convergence. At each iteration t , the deviations of control parameters from their previous configurations are computed as

$$\Delta P_{TX} = |P_{TX}(t) - P_{TX}(t-1)|, \quad (5)$$

$$\Delta OBSS_{PD} = |OBSS_{PD}(t) - OBSS_{PD}(t-1)|. \quad (6)$$

Dynamic thresholds are defined as exponentially decaying functions of the trial index n :

$$r_{TX} = \max(0.125, 5e^{-0.05n}), \quad (7)$$

$$r_{PD} = \max(0.25, 10e^{-0.05n}). \quad (8)$$

A trial is pruned if $\Delta P_{TX} < r_{TX}$ and $\Delta OBSS_{PD} < r_{PD}$, indicating negligible deviation from the prior configuration.

This mechanism suppresses redundant evaluations while maintaining optimization efficiency. **The exponential decay of the thresholds enables a smooth transition from exploration to exploitation**: larger hysteresis margins in early trials encourage broad parameter exploration, whereas progressively tighter bounds in later stages results in fine-grained tuning. Formally,

$$r(n) = ae^{-bn}, \quad (9)$$

where a and b govern the exploration amplitude and convergence rate, respectively, ensuring stable and sample-efficient optimization behavior.

Hysteresis control was not applied to channel bandwidth, as it is a discrete parameter (e.g., 20, 40, 80 MHz). **Applying continuous hysteresis thresholds to such discrete values offers limited benefit and may suppress essential transitions between bandwidth levels**. Instead, bandwidth adaptation was left to the Bayesian optimizer, allowing natural exploration of discrete configurations while maintaining overall stability.

E. MATLAB Evaluation Environment

Validated configurations are sent to the MATLAB RRM engine through the MATLAB–Python API. This engine acts as a highly realistic model that simulates real life conditions, interference from neighboring APs, and Bluetooth coexistence. The simulation is a statistical model that incorporates **stochastic variability** while maintaining system stability, ensuring realistic yet controlled evaluation of network dynamics. The simulation returns metrics such as median throughput (P_{50}), retry rate (P_{95}) and client flag ratio. These results determine whether the configuration is considered stable, unstable, or requires rollback during offline simulations. If the online simulation identifies a configuration as stable and suitable, the system executes a validation run lasting 10 minutes under that configuration and records the average values of P_{50} throughput and P_{95} retry rate observed throughout this period, providing a reliable estimate of steady-state network performance.

F. Constraint Management

Constraint management in the implemented Bayesian optimization framework was achieved using Optuna's Gaussian Process (GP) Sampler with a custom constraint function, `optuna_constraints()`. The GP Sampler inherently supports constraint-aware optimization by incorporating the constraint outputs into its acquisition process, thereby penalizing infeasible trials and promoting exploration within feasible regions.

The enforced system-level constraints were defined as:

$$RP_{95} \leq 8\% \quad (9)$$

$$(P_{TX} + G_{ANT}) \leq 30 \text{ dBm} \quad (10)$$

The constraint function computed scaled penalties for deviations from these limits using the coefficients `RETRY_RATE_PENALTY_DB` and `EIRP_RATE_PENALTY_DB`. By integrating these penalties into its surrogate model, the GP Sampler ensured that constraint satisfaction and objective optimization were jointly achieved, maintaining both efficiency and regulatory compliance throughout the optimization process.

G. Client Feedback, Cool-Off, and IEEE-Aware A/B Toggle

To ensure network stability and consistent performance under dynamic conditions, the system implements a client feedback mechanism. Each client continuously monitors its **throughput** and **retry rate**, raising a flag whenever throughput falls below 50Mbps or the retry rate exceeds 50%, signaling potential link degradation or congestion. When the fraction of flagged clients exceeds 40% of the total active clients, the controller interprets this as a collective performance anomaly and initiates a **30-minute cool-off period**, during which all adaptive optimization processes are temporarily suspended.

During the cool-off phase, the system reverts to a baseline IEEE-standard configuration through an **A/B toggling strategy**, ensuring smooth transition and operational stability. The system operates in two distinct states: **State A (IEEE Baseline)**, with $P_{TX} = 20$ dBm, $OBSS_PD = -75$ dBm, and $WCH = 40$ MHz; and **State B (Bayesian Optimized)**, with parameters suggested by the Bayesian optimization engine. If client dissatisfaction or constraint violations are observed, the controller toggles to State A. Once performance stabilizes under the optimized parameters, State B remains active for approximately 8 minutes **before the next configuration is suggested**.

This A/B toggling mechanism acts as a self-correcting safeguard, permitting Bayesian exploration only when it demonstrably improves performance, while ensuring a fallback to a known-safe IEEE configuration when required. By combining real-time client feedback with controlled configuration switching, the system achieves both **proactive fault recovery** and **robust operational reliability**.

If client dissatisfaction or constraint violation is detected, the controller toggles to the IEEE baseline (State A). If performance stabilizes under the optimized parameters, State B remains active and is held for approximately 8 minutes before enforcing a 20-minute cool-off period. This A/B mechanism provides a self-correcting safeguard—allowing Bayesian exploration only when it benefits performance, and reverting to known-safe IEEE configurations otherwise.

H. Change Budget Enforcement

To ensure gradual and stable network adaptation, a **change budget constraint** is imposed on critical configuration parameters. At each optimization step, the maximum allowable variation for a parameter is defined relative to its operational range. Specifically, the per-step limits are given by:

$$\Delta P_{TX} = 0.15 \times (P_{TX}^{\max} - P_{TX}^{\min})$$

$$\Delta OBSS_PD = 0.2 \times (OBSS_PD^{\max} - \text{NoiseFloor})$$

$$\Delta WCH = 20 \text{ MHz}$$

Parameter bounds are dynamically adjusted based on the previous configuration to ensure that new values remain within both the global limits and the allowed per-step variation:

$$P_{TX}^{\min, \text{new}} = \max(P_{TX}^{\min}, P_{TX}^{\text{prev}} - \Delta P_{TX})$$

$$P_{TX}^{\max, \text{new}} = \min(P_{TX}^{\max}, P_{TX}^{\text{prev}} + \Delta P_{TX})$$

$$OBSS_PD^{\min, \text{new}} = \max(OBSS_PD^{\min}, OBSS_PD^{\text{prev}} - \Delta OBSS_PD)$$

$$OBSS_PD^{\max, \text{new}} = \min(OBSS_PD^{\max}, OBSS_PD^{\text{prev}} + \Delta OBSS_PD)$$

By enforcing these **change budget constraints**, the system avoids abrupt parameter shifts that could destabilize the network, while still allowing adaptive exploration within safe and controlled limits. This mechanism ensures both **gradual adaptation** and **operational stability** during Bayesian optimization.

H. 1. DFS Constraints Management

To maintain compliance with **Dynamic Frequency Selection (DFS)** regulations, the optimization framework enforces a fixed primary channel and restricts exploration to permissible bandwidths. The set of allowable bandwidths is dynamically updated in each optimization window based on current radar detections, ensuring that no candidate configuration overlaps with a DFS-protected sub-channel. During each Optuna trial, any bandwidth whose spectral mask intersects a radar-restricted range is rejected through the constraint callback. This prevents the surrogate model from evaluating or proposing DFS-violating configurations and ensures that Bayesian Optimization operates strictly within regulatory bounds.

Overall, this architecture forms a hybrid closed-loop control system that merges the predictive intelligence of Bayesian learning with the operational safety of IEEE-compliant RRM. It continuously alternates between exploration, validation, rollback, and stability maintenance—mirroring the sequence illustrated in Fig. 1.

V. SYSTEM DYNAMICS AND BEHAVIOR

The end-to-end RRM control loop is designed to operate reliably under continuously evolving real-world conditions by integrating learning, safety, and temporal coordination into a unified optimization pipeline. Offline simulations provide the initial surrogate structure for the Gaussian Process (GP) sampler, allowing Bayesian Optimization (BO) to begin online exploration with an informed prior rather than a cold start. During deployment, the system operates in discrete time slots (e.g., 3-hour windows), within which the GP sampler proposes candidate configurations that respect all constraint layers, including DFS-safe bandwidth sets, change-budget limits, and parameter guardrails. Hysteresis control and cooldown logic suppress unnecessary oscillations by preventing rapid parameter flips when performance metrics fluctuate near decision boundaries. Real-time client feedback—such as throughput distribution, retry rate, and SINR statistics—drives drift detection and determines whether the BO search should be reactivated or remain idle. Prior to system-wide rollout, an A/B toggle mechanism validates each candidate configuration against the current baseline, ensuring measurable improvement

without inducing transient instability. These components collectively enable the controller to remain robust under dynamic environmental conditions—including client mobility, emerging interferers, and shifting traffic load—while continuously refining the RRM policy in a safe and constraint-aware manner.

VI. RESULTS AND ANALYSIS

This section summarizes the results obtained from the MATLAB-integrated simulation and BO optimization runs.

Metric	Cisco	Optimized	Improvement. (%)
P50 Throughput (Mbps)	159.8	171.43	7.28
P95 Retry Rate (%)	96.8	0.06	99.93
Client Complaint Rate (%)	0.17	0.10	50.82

TABLE I: comparison of performance metrics between baseline and bo-optimized configurations.

A. Visual Comparisons

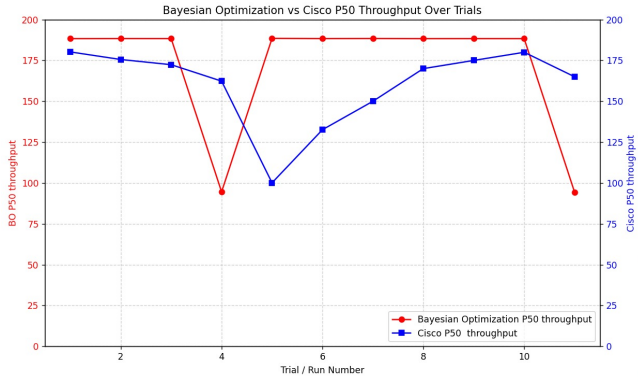


Fig. 2: Bayesian and Cisco Throughput Comparison

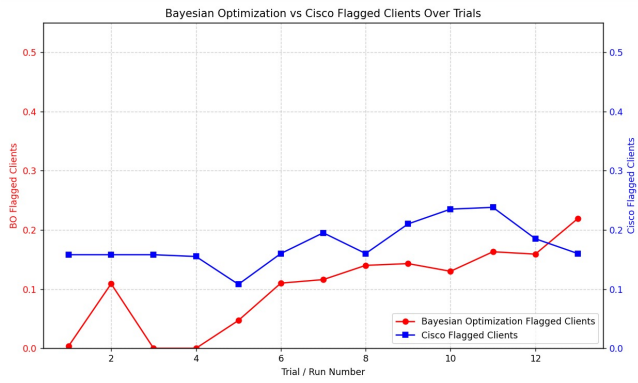


Fig. 3: No of clients Flagged (Clients Complaints).

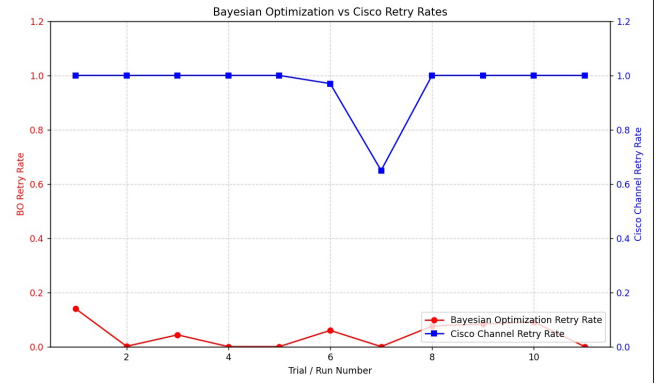


Fig. 4: P95 Retry Rate Comparison

The Bayesian Optimizer consistently achieves higher throughput than Cisco's default strategy. Even when small drops in performance occur, its rollback mechanism quickly returns the system to the last stable setting, preventing long periods of degraded performance. Cisco's approach, on the other hand, shows an early dip and only recovers slowly, with improvements that are less steady over time.

Client flags (or complaints) also remain lower when using the Bayesian method, suggesting a more reliable and user-friendly network experience.

In addition, the Bayesian Optimizer maintains a lower P95 retry rate, meaning clients face fewer retransmissions and more stable link conditions. Cisco's strategy tends to produce more variation and generally higher retry rates.

Overall, these observations highlight that the Bayesian optimization strategy converges faster, behaves more robustly under changing conditions, and offers a more consistent quality of experience for users.

Furthermore, the P95 retry rate remains lower for the Bayesian Optimizer, showing that clients experience fewer retransmissions and more stable link conditions, whereas Cisco's strategy results in more fluctuating and often higher retry rates. Together, these results demonstrate the superior convergence behavior, robustness, and user-centric performance of the Bayesian optimization strategy.

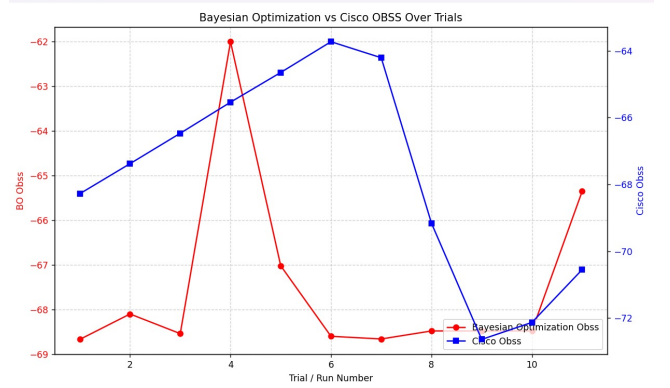


Fig. 5: OBSS threshold comparison

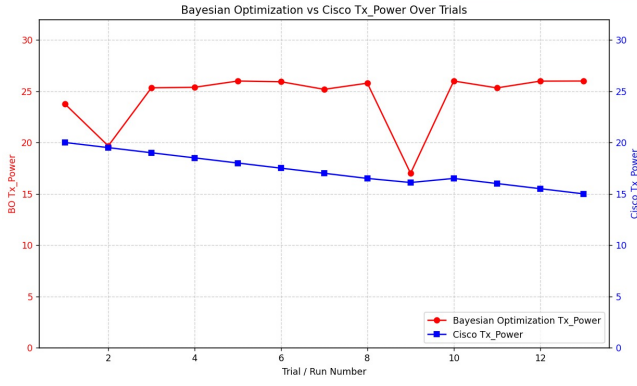


Fig. 6: Transmission Power

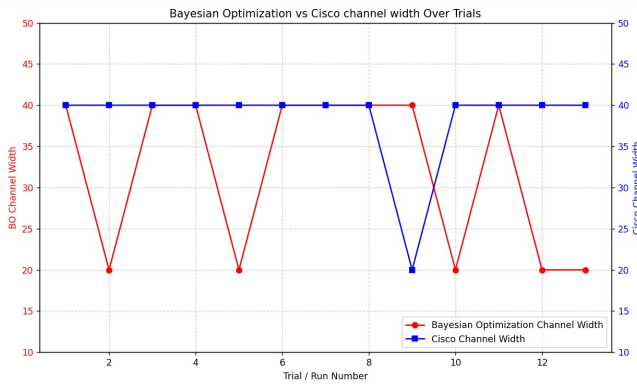


Fig. 7: Channel Width

The input parameters are chosen according to the proposed BO strategy. However Cisco's strategy tries to reduce the transmission power if the client flag rate is less to reduce power consumption whereas Bayesian Optimiser strategy completely focuses on optimising throughput without considering this parameter. Hence Cisco's Strategy consistently uses less transmission power.

VII. DISCUSSION: WHY HYBRID BO BEATS STATIC HEURISTICS

Traditional RRM algorithms, such as those used by Cisco, rely on environment-sensing thresholds and fixed reactive triggers. Their algorithms converge slowly fundamentally lack adaptivity to handle the complex, fluctuating interference patterns of modern Wi-Fi environments.

The proposed Hybrid Bayesian Optimization (BO) framework overcomes these limitations by generalizing across unknown interference patterns and learning the function mapping the control parameters $\mathbf{x} = (P_{TX}, OBSS_{PD}, W_{CH})$ to throughput. Once trained, BO can re-optimize in minimal iterations, maintaining real-time adaptability with minimal oscillations.

Key Advantages of the BO Approach::

- **Superior Performance:** The BO strategy consistently achieves higher median throughput (P_{50}) and significantly lower P95 Retry Rates compared to the baseline approach.

- **Robustness and Stability:** The system's architecture, featuring a rollback mechanism and hysteresis control, ensures that the system quickly recovers from performance dips, behaving more robustly than Cisco's approach, which recovers slowly.
- **Constraint-Awareness:** The framework incorporates constraint management to ensure compliance with critical limits, such as the maximum EIRP (≤ 30 dBm) and a P95 Retry Rate constraint of $\leq 8\%$
- **Client-Centric Improvement:** Client complaints remain lower when using the Bayesian method, suggesting a more reliable and user-friendly experience.

VIII. CONCLUSION

This hybrid RRM approach unites **IEEE-defined stability with data-driven optimization**. By embedding Bayesian learning inside a constraint-governed, budget-aware architecture, it achieves a fine balance between agility and reliability.

The system's success is defined by its architectural safeguards:

- **Operational Safety:** The inclusion of the **A/B toggle** strategy ensures field safety by providing a self-correcting safeguard and a fallback to a known-safe IEEE baseline configuration when constraint violations or client dissatisfaction are detected.
- **Continuous Improvement:** The architecture is a hybrid closed-loop control system that continuously alternates between exploration, validation, rollback, and stability maintenance.
- **Deployability:** By adhering to regulatory and stability constraints, the system is deployable in real enterprise WLANs while maintaining compliance and continuous self-improvement.

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